Course Outline for CSE-356

Part A

1. Course Code: CSE-356

2. Course Title: Software Engineering (Sessional)

3. Course Type: Core Course

4.Level/ Term: Level: 3 Term: II

5. Academic Session: 2020-21

6. Course Teacher: Mir. Md. Saki Kowsar, Assistant Professor, Dept. of CSE, CUET
Md. Sabir Hossain, Assistant Professor, Dept. of CSE, CUET
Ashim Dey, Lecturer, Dept. of CSE, CUET

7. Prerequisite(s): None

8. Credits: 0.75

9. Contact Hours: (3 hours of lab work per two week)

10. Total Marks: 75

11. Rational of the Course:

This course will cover the basics of design, deployment and maintenance of a medium scale software development project. This course is important to understand the modern software models, techniques and technologies to bring out innovative and novelistic solution. This a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is

Course Content:

Sessional based on the following topics:

Concepts of software engineering; Software engineering paradigms; Different phases of software, Synthesis vs. iterative design Top-down and bottom-up design; Different design tools; Structured and non-Structured programming, Data-directed design techniques, Modular design, Design of automatic, redundant and defensive programs; Influences of languages in design process Concepts of complexity measures; COCOMO model; Tree model; PNR curve, Statistical model; Zipf's laws and their application in computer

languages; Halstead program length formula; Graphical analysis for complexity measures, Memory requirements analysis, Processing time analysis, Testing philosophy; Test methods; Debugging Verification, Validation and certification, Choice of test data; Simulator; Arthor Laemmel's scheme; Concepts of software reliability and availability: Software repair, downtime, error and faults, specification and correction; New error generation hypothesis, Estimating number of bugs in computer program, Reliability models, Availability models; Quality assurance; Quality measures Different cost estimation models and their comparisons, Software maintenance; Maintenance-cost models; Growth dynamic models; Documentation; Software project organization; Management and communication skills.

12. Course Objectives:

(a) Analyze complex real-world problems and devise efficient software-based solutions.
(b) Apply modern software models, techniques and technologies to bring out innovative and novelistic solution.
(c) Gain experience of doing design, deployment and maintenance of a medium scale software development project.
(d) Demonstrate effective and modern working strategies by possessing the leadership skills as an individual or contributing to the team development.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

No.	Course Learning Outcomes (CLOs)	Bloom's Level (Optional)
CLO1	**Prepare** a project management plan by selecting a lifecycle model and **determine** milestones on basis of risk and cost analysis.	C3, C5, A4
CLO2	**State** the requirements through an effective communication with various stakeholders and represent them using UML diagrams	C1, P2
CLO3	**Identify** a software architectural model that describes the software's top level structure and identifies its components.	C1, C4, A1, A4, P1
CLO4	**Develop** the software based on the required specification and **evaluate** the performance, maintenance and quality requirements of the developed software through testing**.**	C5
CLO5	**Work** as an individual and as part of a multidisciplinary team to develop and deliver quality software by creating collaborative and inclusive environment.	A3

b) Mapping of CLOs with PLOs

No.	CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
1	CLO1											X	
2	CLO2										X		
3	CLO3		X										
4	CLO4			X	X								
5	CLO5									X			

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

Course Plan

`	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Introduction to Software Engineering Lab.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-1
Week - 02	Identification of a complex real-world problem and proposing an efficient software-based solution.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-1
Week - 03	Specification of the requirements of the stakeholders of the proposed SW.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 04	Illustration of the SW architectural model and its major	• Lecture on theatrical background and design principle	Lab Performance Report	• CLO-3

	components, their relationships	• Hands on demonstration on the implementation • Group Discussion		
Week - 05	Implementation of the proposed SW and testing the SW against user requirements.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-4
Week - 06	Project Submission and Viva-voce	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO5
Week - 07	Quiz			

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Quizzes	15%
Viva-voce	15%
Class performance including reports	60%
Attendance	10%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 70%
b) Summative: 30%

3) Make-up Procedures:

- Course teacher may arrange for makeup lab schedule if necessary.

Part D

16. Learning Materials

1) Recommended Readings

- Software Engineering by Martin L. Shooman (Int. Student Edition), McGraw Hill.

2) Supplementary Readings

- Software Engineering by Roger S. Pressman (8th Edition), McGraw Hill.